Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	**Aim: To designand implementdataselection and routing systemsusing FPGA.**	
Experiment No: 4	**Date:**	**Enrolment No: 92301733013**

Aim: **To design and implement data selection and routing systems using FPGA.**

Software required: Quartus II 13.0sp1

Hardware required: Altera Cyclone II FPGA Board

Theory:

Data selection and routing are essential operations in digital systems. They allow digital data to be selected from multiple inputs and routed to a desired destination. In FPGA-based systems, these functions are implemented using combinational logic such as Multiplexers (MUX) and Demultiplexers (DEMUX).

The FPGA (Field Programmable Gate Array) contains configurable logic blocks (CLBs), lookup tables (LUTs), programmable interconnections, and I/O blocks. By writing Verilog/VHDL code and downloading it into the FPGA, the hardware can be configured to perform different routing and selection operations.

Data Selection (Multiplexer)

A Multiplexer (MUX) is a digital circuit that selects one input from several inputs and forwards it to a single output.

- Also called a Data Selector.
- Controlled by selection lines.
- Used in communication systems, processors, ALUs, and memory selection.

Example:

- 4 Inputs → I0, I1, I2, I3
- 2 Select Lines → S1, S0
- 1 Output → Y

The selected input depends on the binary value of the select lines.

Data Routing (Demultiplexer)

A Demultiplexer (DEMUX) performs the opposite function.

It takes

- One input
- Routes it to one of several outputs

using selection lines.

Applications include

- Memory addressing
- Signal distribution
- Communication systems
- FPGA routing

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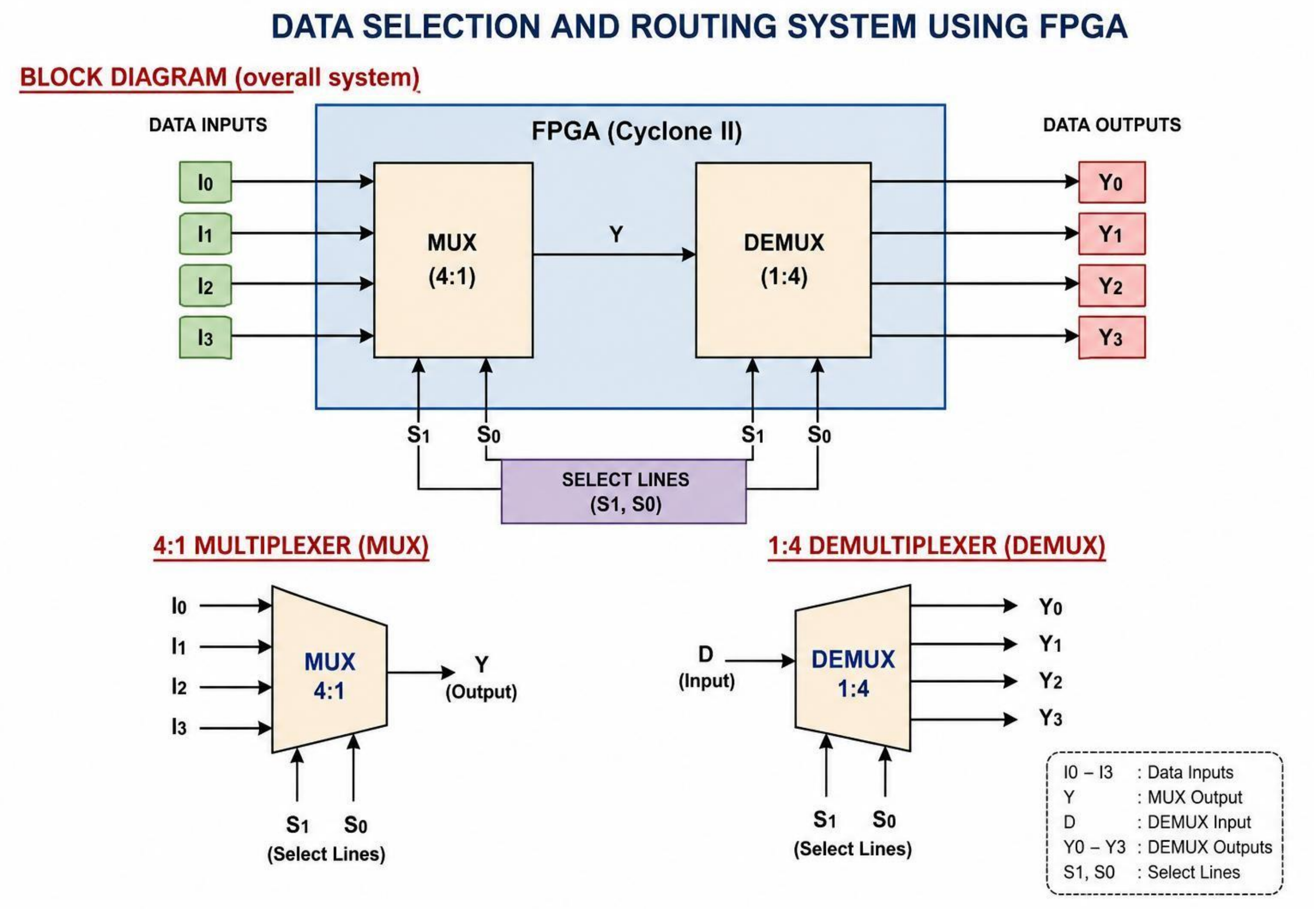
Block diagram:



Truth Table:

Truth Table of 4:1 Multiplexer

S1	S0	Output Y
0	0	I0
0	1	I1
1	0	I2
1	1	I3

Truth Table of 1:4 Demultiplexer

Assume Input = 1

S1	S0	Y0	Y1	Y2	Y3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

If Input = 0, then all outputs remain **0**.

Boolean Expression:

$$Y = I_0\overline{S_1}\,\overline{S_0} + I_1\overline{S_1}S_0 + I_2S_1\overline{S_0} + I_3S_1S_0$$

$$Y_0 = D\overline{S_1}\,\overline{S_0}$$

$$Y_1 = D\overline{S_1}S_0$$

$$Y_2 = DS_1\overline{S_0}$$

$$Y_3 = DS_1S_0$$

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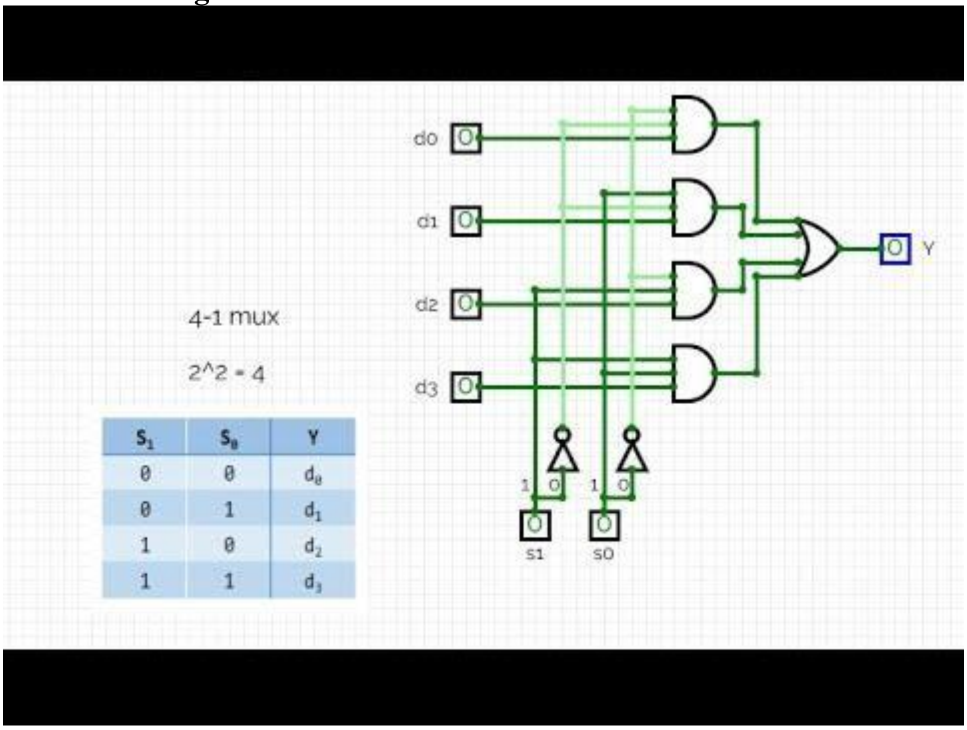
Circuit Diagram:



Verilog Code:

```
module MUX4x1(
    input wire a, b, c, d,
    input wire s0, s1,
    output reg y
);

always @(*) begin
    case ({s1,s0})
        2'b00: y = a;
        2'b01: y = b;
        2'b10: y = c;
        2'b11: y = d;
    endcase
end

endmodule
```

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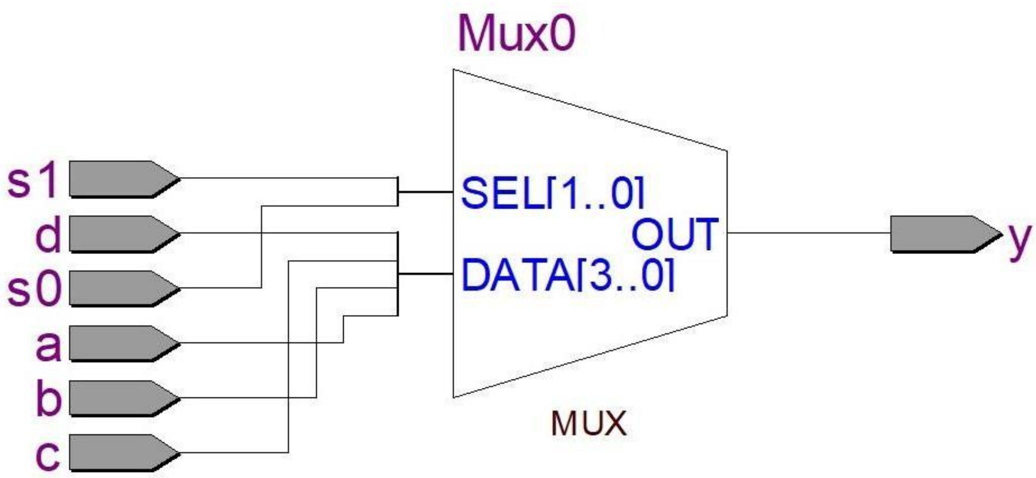
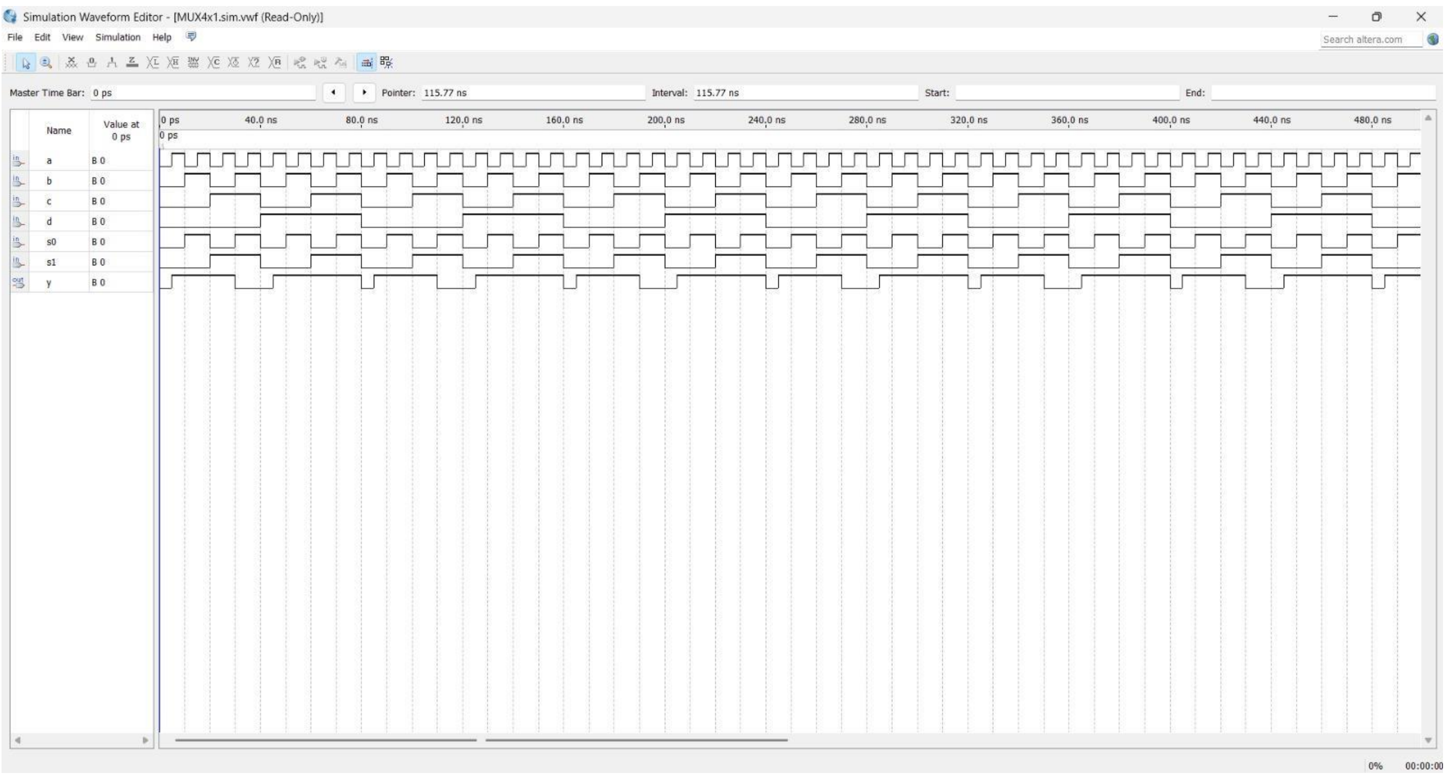
<u>Simulation Results:</u>

RTL Schematic View:-



Simulation Waveform:-



FPGA Implementation with Pin Planning Details:-

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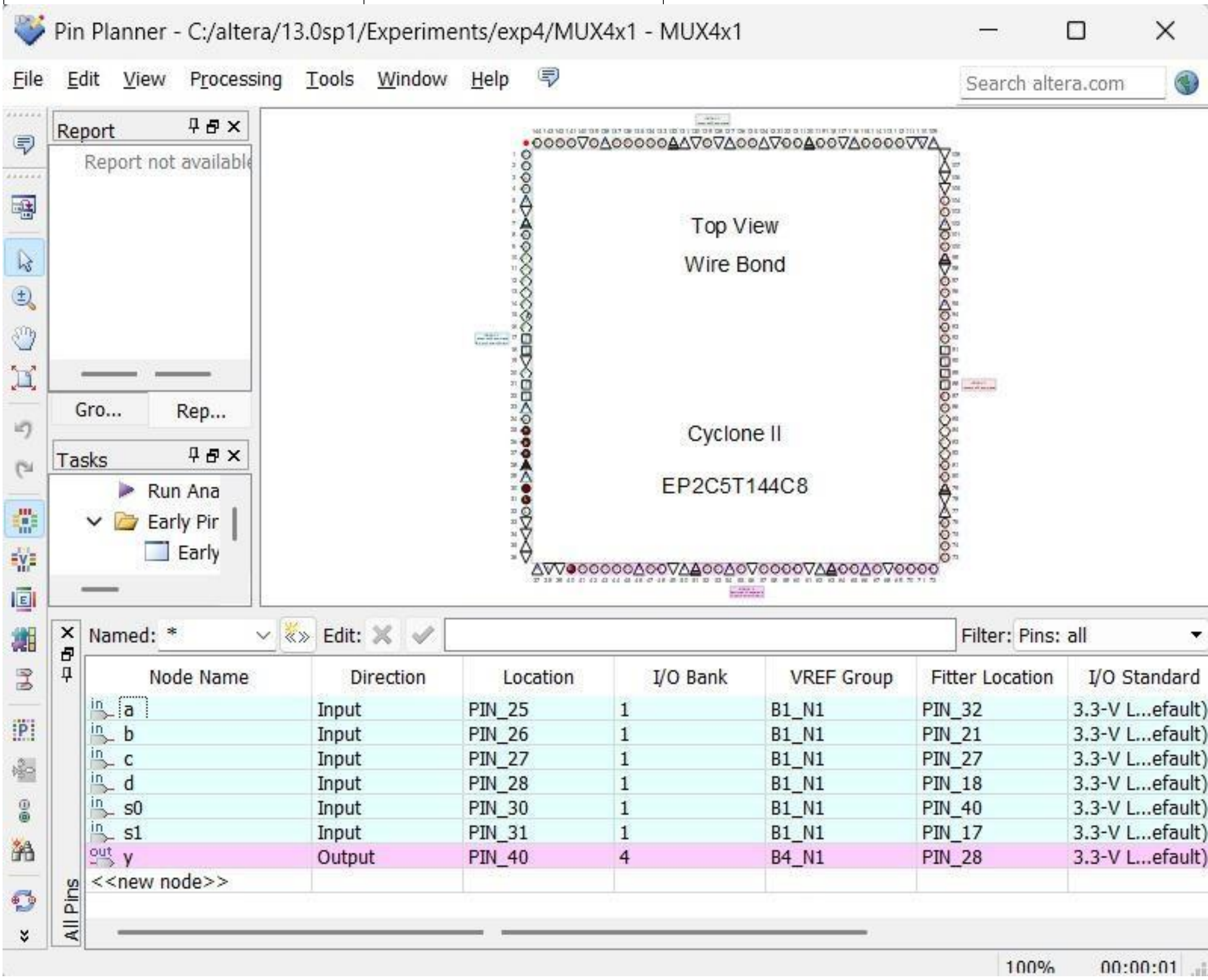
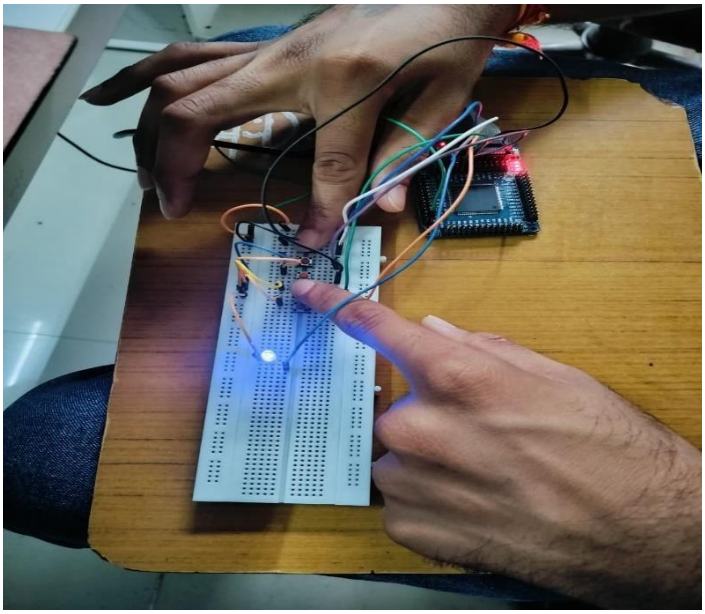


Conclusion

The experiment was successfully completed by designing and implementing a 4:1 Multiplexer (MUX) and a 1:4 Demultiplexer (DEMUX) using Verilog HDL on the Altera Cyclone II FPGA. The MUX was used to select one of four input signals based on the given select lines, while the DEMUX was used to route a single input signal to the selected output.

The designs were verified through simulation, and the obtained results matched the expected behavior of both circuits. This experiment helped us understand how combinational logic circuits can be implemented and tested on an FPGA. It also provided practical experience with Verilog HDL, FPGA programming, simulation, and digital circuit design, while demonstrating the flexibility and reconfigurability offered by FPGA technology.



Post Lab Exercise:

Post Lab Exercise

1. What is the function of a multiplexer (MUX)?

A Multiplexer (MUX) is a combinational logic circuit that selects one input from several available inputs and sends the selected signal to a single output. The selection is controlled by the select lines. A MUX is also commonly called a Data Selector.

2. How many select lines are required for a 4:1 multiplexer?

A 4:1 Multiplexer requires 2 select lines, usually represented as S1 and S0.

The number of select lines can be calculated using:

Number of Select Lines = $\log_2$(Number of Inputs)

Therefore:

$\log_2(4) = 2$

These two select lines determine which one of the four inputs (I0, I1, I2, or I3) is connected to the output.

3. How many outputs does a 1:8 demultiplexer have?

A 1:8 Demultiplexer has 8 output lines, represented as Y0 to Y7. It takes a single input signal and sends it to one of the eight outputs based on the values of the select lines.

4. What determines the active output in a demultiplexer?

The select lines determine which output of the demultiplexer becomes active. Depending on the binary value of the select lines, the input signal is directed to the corresponding output, while the remaining outputs stay inactive.

5. What is the main advantage of implementing MUX and DEMUX on an FPGA?

The main advantage is the flexibility and reconfigurability of the FPGA. MUX and DEMUX circuits can be designed, modified, and tested by simply changing the Verilog or VHDL code, without having to change the physical hardware. This makes development faster and allows digital circuits to be tested and improved easily.

6. Why is simulation performed before programming the FPGA?

Simulation is carried out to check whether the Verilog design works correctly before it is programmed onto the actual FPGA. It helps identify logical errors, verify that the outputs match the expected results, and reduce debugging time during hardware testing. RTL schematics and waveforms generated during simulation also help in understanding and verifying the circuit before implementation.

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